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Dr. Ahmet M. Elbir
Associate Editor
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Dear Dr. Elbir,

We submit a new manuscript, “A Closed-Loop Subspace-RFS Framework for Multi-Target DOA Tracking,” which is a substantially revised version of our earlier submission T-SP-35123-2026 (“... via COP-RFS”). In accordance with IEEE SPS Policy 6.16 on the handling of rejected papers, we wish to make clear that this is a *substantially different* manuscript: it has a new central thesis, two new theoretical results that were absent from the prior version, corrected derivations, and a completely re-run experimental evaluation. A detailed point-by-point response to all prior reviewer comments accompanies this submission.

The prior reviews converged on one central concern: the work read as an *integration* of established components (the authors’ COP front-end and a GM-PHD tracker) rather than a fundamentally new contribution. We have addressed this at the root by changing what the paper is *about*:

- **The contribution is now the closed-loop coupling itself, established as a front-end-agnostic principle.** We prove (i) a posterior Cramér–Rao bound for the complete closed-loop estimator–tracker (Theorem 1), showing the temporal prior restores identifiability exactly in the underdetermined regime where the data-only bound diverges; and (ii) that the tracker-aided subspace refinement is minimum-variance on the Grassmann manifold under an explicit, gate-enforced condition (Theorem 2). These are new results, not present in the rejected version.
- **We demonstrate the principle on two different front-ends** (the higher-order-cumulant COP and the covariance-based MUSIC), showing the closed loop benefits both—lowering localization error by 56% (COP) and 44% (MUSIC) at low SNR—evidence that the contribution is general rather than a specific algorithm pairing.
- **Every flagged derivation has been corrected:** the effective sample size (Kish), the basis-invariant subspace refinement (replacing the gauge-dependent interpolation), the velocity gate (removing double-counting), the cumulant-model assumptions, and the cumulant-noise covariance. The unsupported “exact” CRLB and the heuristic convergence bound have been removed; the back-end is renamed a track-oriented PHD with an honest statement of its relation to the exact recursion.
- **The experiments are re-run with up to 500 Monte-Carlo trials and 95% confidence intervals**, with a closed-loop-MUSIC baseline, a 2×2 (front-end $\times$ loop) ablation, and a tracking back-end ablation against the LMB and δ -GLMB labeled-RFS filters. We now explicitly *delineate the operating regime* in which the higher-order front-end is advantageous, and we have removed all overclaims (“tracks all,” “substantially outperforms”).

Crucially, the revised paper now *quantifies* the value of closing the loop—a concrete gain in every regime where the temporal prior is informative (restored identifiability, -56% low-SNR localization error, -15% identity switches under target motion), with a provable no-harm guarantee elsewhere. We believe these changes squarely address the reviewers’ concerns about novelty positioning, theoretical rigor, and experimental support, and that the resulting paper makes a clear theoretical contribution appropriate for the Transactions. Thank you for considering our work.

Sincerely,

Jin Ho Choi